KCE PATHWAYS · GRADES 4–5 · UNIT 4: SHAPES AND SPACE · WEEK 19 · MATH

The Peach Packing House

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Pre-Lesson Warm-Up: How Much Fits Inside?

Makena is spending Saturday at a peach packing house in Fort Valley, Georgia. Warm up before the lesson, and show your work. Some problems may feel new, and that is okay!

1. Review (Week 18): Find each missing angle. a) Two angles form a straight line. One is 125°. What is the other? b) A right angle is split into two parts. One part is 34°. What is the other?

2. A box is filled with unit cubes. The bottom layer has 4 rows of 5 cubes. The box is 2 layers tall. How many cubes fill the box?

3. The bottom of a peach box is a rectangle 6 inches long and 3 inches wide. What is the area of the bottom?

4. Which unit would you use for the amount of space INSIDE a box?

   inches | square inches | cubic inches

5. Draw a box shape (a rectangular prism). Label its length, width, and height.

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POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Post-Lesson Challenge: Packing Day

Write the formula you use, show your work, and label answers with cubic units.

1. a) A wooden peach bin is 9 feet long, 6 feet wide, and 5 feet tall. What is its volume? b) A box has a base area of 42 square inches and a height of 8 inches. What is its volume?

2. A loading platform is made of two prisms. Prism A is 10 feet long, 4 feet wide, and 2 feet tall. Prism B is 5 feet long, 4 feet wide, and 3 feet tall. What is the total volume?

3. How many crates shaped like cubes with 2-foot edges fit in a space that is 12 feet long, 6 feet wide, and 4 feet tall?

   24 crates | 36 crates | 72 crates | 288 crates

4. Find the mistake: Wendell says a cube with 4-foot edges has a volume of 16 cubic feet, because 4 × 4 = 16. In two or three complete sentences, explain his mistake and give the correct volume.

★ **Challenge:** Uncle Thaddeus needs a box that holds exactly 600 cubic inches. Its base is 12 inches long and 10 inches wide. How tall must it be? If he doubles only the height, what is the new volume?

Looking Ahead: Week 20: Unit 4 Checkpoint. Review unit fraction division, decimal division, angles, and volume.